

COLLISION REPAIR INFORMATION BULLETIN



SUBJECT:	SPOT WELD REMOVAL METHODS AND TOOLS		
BULLETIN NO.:	TS-TCI-CRIB-2028	DATE ISSUED:	May, 2013
MODELS:	All Toyota & Scion Models	SECTION:	Welding

The removal and replacement of spot welds during unibody panel replacement can be performed quickly and efficiently by the following methods:

- Drill-Through Method
- Top-Drill Method
- Hole-Saw Method

This bulletin sets out the three spot weld removal methods and recommended tools to help you decide which method to use during a collision repair.

1. Drill-Through Method:

This method can be used when a replacement panel is welded under or sandwiched between existing panels (see illustration A).

Use the drill-through method when:

- The replacement panel is behind or beneath an existing panel.
- The replacement panel is between existing panels and the welding access is from either the front or the back side. Center punch the weld first and then use the appropriate size drill for the panel thickness and weld diameter. Drill size is typically 6 – 8 mm (0.24 – 0.31 in.).

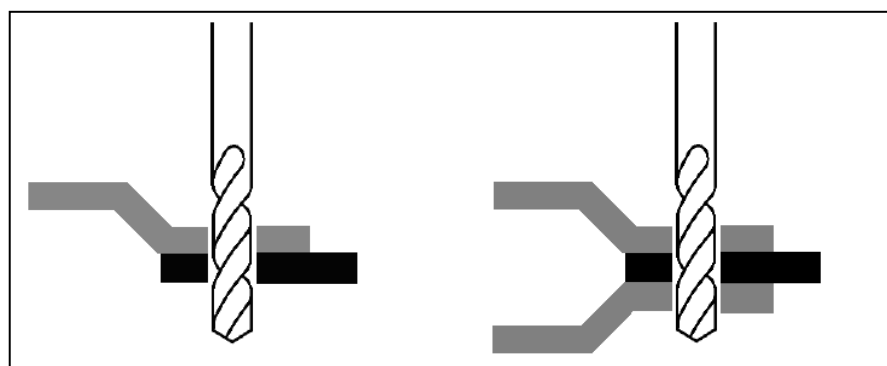


Illustration A

NOTE: Issue of this Collision Repair Information Bulletin by Toyota Canada Inc. is to ensure that proper repair information is available. It does not constitute an authorization for repair. PLEASE ROUTE THIS BULLETIN TO YOUR COLLISION REPAIR CENTER MANAGER AND COLLISION & REFINISH TECHNICIANS.



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2. Top-Drill Method:

This method is used when the top or middle panel is removed without damaging the panel beneath. The spot weld is center punched first and then drilled until the measured depth of the top sheet steel is reached. This will remove the spot weld without unnecessarily removing material from the panel below. It also provides the correct diameter hole for MIG welding. Use a 6 – 8 mm (0.24 – 0.31 in.) drill. See illustration B and C.

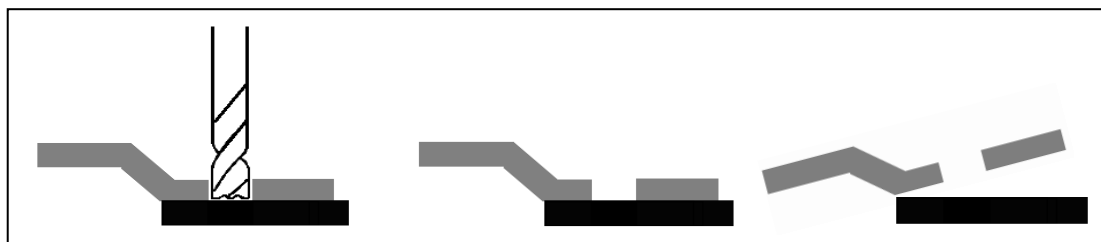


Illustration B – To Remove Top Panel Only

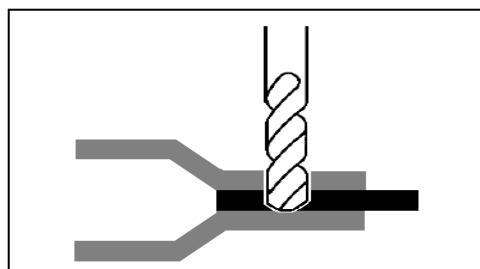


Illustration C – To Remove Middle Panel

3. Hole-Saw Method:

This method uses a hole saw spot weld remover. Essentially it is the same as the top drill method except the tool used has an additional feature. This tool has replaceable cutters in various sizes (ranging from 0.635 mm or 1/4" to 11.1125 mm or 7/16"). To use a small hole saw type spot weld remover, first center punch the spot weld (see illustration D). Next, drill the spot as you normally would. The small nib of metal left after the panel has been removed is eliminated by using a small metal grinding wheel (see illustration E).

TIP: Using a variable speed electric drill or an air drill with controllable air pressure and slow speed may help extend the life of the bit.

NOTICE: Always follow any required safety procedures and use all necessary protective clothing or equipment when performing the operations described in the bulletin.

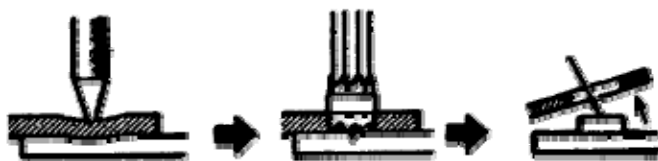


Illustration D



Illustration E



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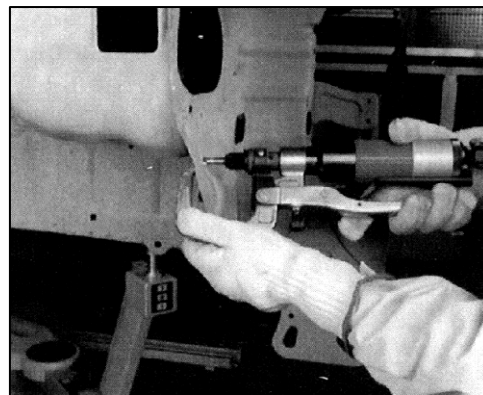
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4. Tools:

The pictures below show two different types of spot weld removal tools that can support the drill-through and top-drill methods. Using the proper tools simplifies the spot weld removal process and can help improve efficiency of the operation (see illustration E).



Types of Spot Weld Drills



Typical Spot Weld Removal Use

Illustration E

5. Spot Weld Locations and Removal Methods:

The Repair Manuals for Collision Damage for the specific Toyota/Scion model indicates the precise spot weld locations and drilling methods. See example provided in Illustration F below.

*1 Position of panels that are removed		
*2 Weld points		
SYMBOLS	MEANING	ILLUSTRATION
▲	Remove Weld Points	
○	Position of Panel Being Replaced is on Outside	
M	Position of Panel Being Replaced is in Center	
I	Position of Panel Being Replaced is on Inside	

Illustration F

Additional sources of structural repair information are available in the Collision Damage Repair Manuals for each Toyota/Scion model and should be consulted before embarking on a collision repair. Always refer to the model specific Repair Manual.



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In partnership with Würth Canada, Toyota Canada Inc. (TCI) recommends and offers a complete line-up of body shop supplies, adhesives, fasteners, abrasives, connectors, safety supplies, tools, etc., that are available through the Würth Body Shop Products Program. Toyota and Scion Dealers can order all of Würth's products through this program. Dealers can access the program details in the Infostream website portal at <http://infostream.toyota.ca/>. For independently operated collision centres interested in these products, please contact your local Toyota and Scion Dealer.

Specifically, in regard to this bulletin, Toyota Canada Inc. recommends the following Würth products:

Würth Product Name	Art. No.	Illustration
Drill Bit (1/4") or Drill Bit (5/16")	627.14 or 627.516	
Spot Weld Drill Bit (6mm long life tin coated) or Spot Weld Drill Bit (8mm long life tin coated)	710.0066 or 710.0088	
Spot Weld Drill Bit (6mm HSS) or Spot Weld Drill Bit (8mm HSS)	710.6 or 10.8	
2 Piece Spring Spot Weld Bit (8mm)	698.1	
3" Cutting Disc	670.763	
Safety Glasses, "Trendus" Clear	899.103132	